

IERG4340 FINAL PRESENTATION

GPT-Powered Paper Trading Bot

A real-time, risk-aware assistant for disciplined paper trading decisions.

Live + Cached

Resilient data mode
switching

Multi-Market

Stocks, options, forex, crypto

Risk Panels

Drawdown-aware controls

Automation

Logs, screens, and backtests

The Problem

SIGNAL OVERLOAD

Traders cannot reliably process multiple market streams fast enough.

RISK DRIFT

Manual workflows delay defensive actions when volatility spikes.

EXECUTION FRICTION

Great ideas fail when tooling lacks integrated execution and feedback loops.

Market Evidence and Scale

THE MARKET IS MASSIVE AND INCREASINGLY MOBILE-FIRST

8.0B

GLOBAL POPULATION

UN milestone reached in Nov 2022.

58%

MOBILE INTERNET CONNECTED

GSMA SoMIC 2025 global view for 2024.

\$7.5T

DAILY FX TURNOVER

BIS Triennial Survey 2022, OTC market.

62%

US STOCK OWNERSHIP

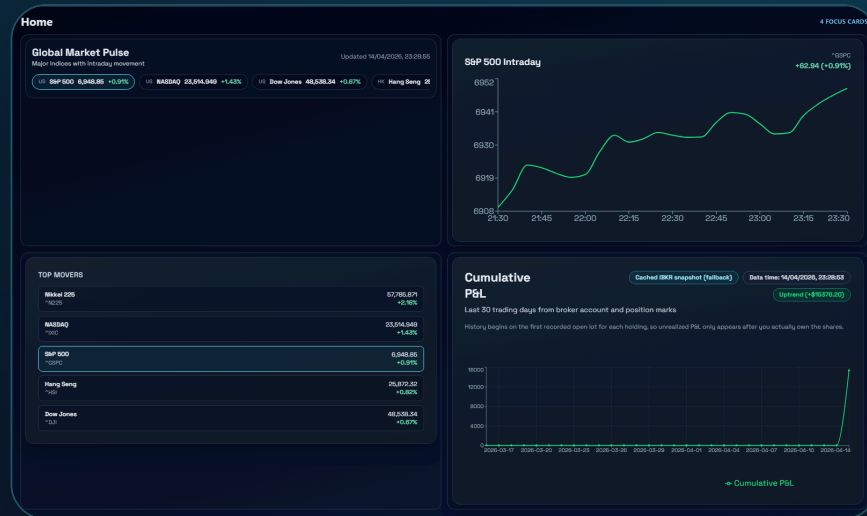
Gallup 2025 update, broad investor participation.

Sources used via MCP web fetch: United Nations Population page, GSMA SoMIC 2025, BIS FX Triennial Survey 2022, Gallup stock ownership 2025.

Solution Overview

UNIFIED COMMAND CENTER FOR THE FULL TRADING LOOP

- Discover opportunities in Trade Desk
- Validate risk before every action
- Track performance, logs, and bot behavior in real time
- Keep transparency under data delays



Trade Desk

FAST SIGNAL TRIAGE AND CONFIGURABLE
STRATEGY LOGIC

- Opportunity feed and context in one place
- Strategy tab for rule-level adjustments
- Built for rapid daily decision cycles

Trade Desk

2 FOCUS CARDS

My Scanner Presets

Preset Name

My Momentum Setup

Save Current Setup

No saved presets yet. Save your current setup to reuse it later.

Quick Scanner Chips

Breakout Movers

High relative volume and momentum names.

Value Setups

Lower valuation candidates with decent liquidity.

Large-Cap Quality

Bigger companies with analyst coverage and stable screens.

Analyst Favorites

Names with stronger analyst participation.

DCF Upside

Candidates where valuation model is available.

Trade Desk - Strategy

3 FOCUS CARDS

Select Strategy

Blowup Stocks

Covered Calls

Forex

Current Market Regime

VIX: 16.31

Bull

Trending up, low VIX - favor growth strategies

Entry & Exit Rules

Guided Presets (Recommended for non-experts)

Conservative

Lower risk, higher risk, prioritize capital protection.

Balanced

Moderate risk/reward and medium holding horizon.

Aggressive

Higher return tolerance for stronger upside attempts.

ENTRY RULES

MAX PE RATIO

30

MAX RELATIVE VOLUME

2

MIN SCORE

50

EXIT RULES

MAX HOLDING DAYS

30

How long to hold before forcing an exit if target/stop are not hit.

PROFIT TARGET PCT

5

STOP LOSS (%)

2

Maximum tolerated loss per position before auto exit.

RISK CONTROLS

MAX HOLDING DAYS

30

How long to hold before forcing an exit if target/stop are not hit.

MAX POSITION SIZE (%)

5

Largest single position allocation.

TAKE PROFIT (%)

5

Profit target per position before taking gains.

MAX DAILY LOSS (%)

1

Maximum total daily loss before risk-off behavior.

STOP LOSS (%)

2

Maximum tolerated loss per position before auto exit.

TRAILING STOP (%)

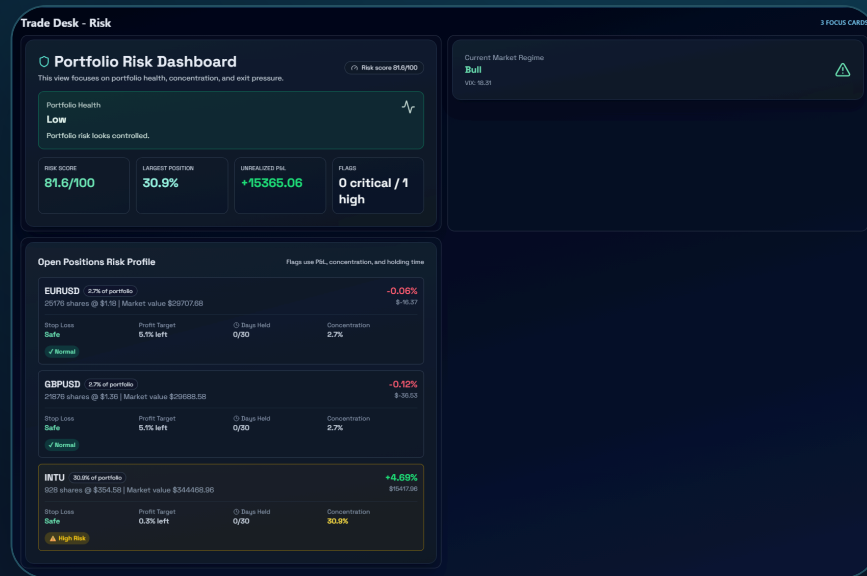
Trailing stop to protect gains as price rises.

Save Strategy Config

Risk-First Execution

CONTROLS THAT PREVENT WIN-SMALL, LOSE-BIG BEHAVIOR

- Dedicated risk panel linked to positions
- Guardrails for drawdown-sensitive operation
- Clear visibility before execution



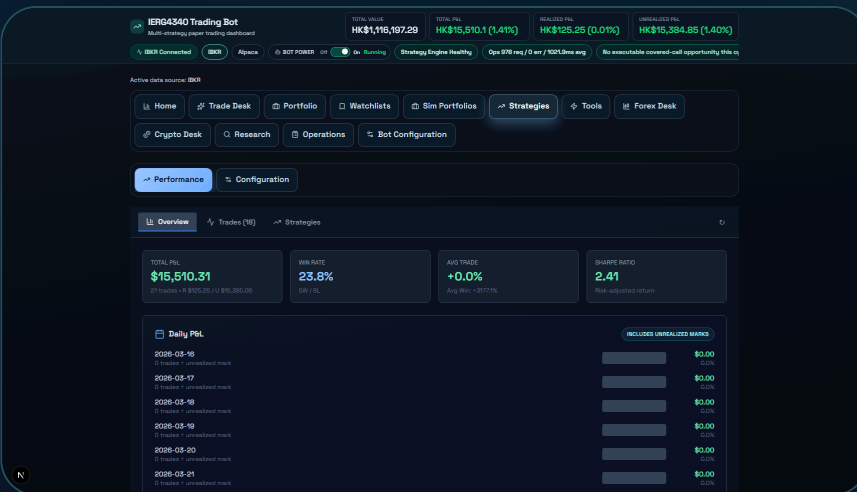
Portfolio and Strategy Intelligence

MEASURE WHAT WORKS, ADJUST WHAT DOES NOT

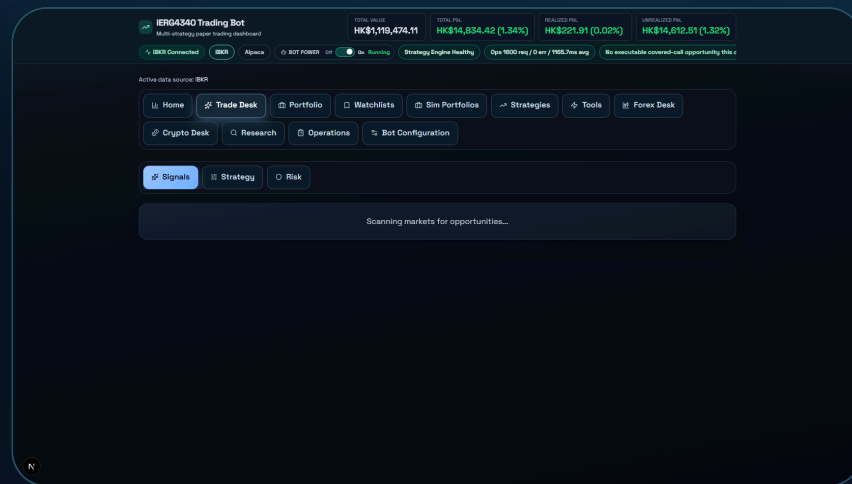
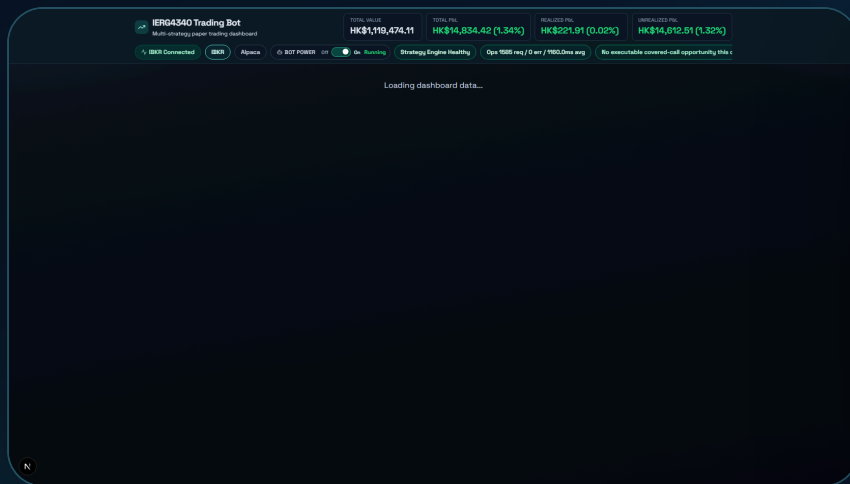
- Portfolio PnL and exposure visibility
- Strategy performance tracking
- Strategy configuration for iterative tuning

Portfolio 1 FOCUS CARDS

Symbol	Qty	Avg Price	Last Price	Market Value	Unrealized P&L	Return
EURUSD	25,176	\$1.18	\$1.18	\$29,706.80	-\$17.26	-0.06%
GBPUSD	21,876	\$1.39	\$1.39	\$30,887.92	-\$37.18	-0.13%
INTU	928	\$354.58	\$371.18	\$344,455.04	+\$15,404.04	+4.68%



Full Web Layout (Desktop)



Full Web Layout (More Views)

IERG4340 Trading Bot

Multi-strategy paper trading dashboard

TOTAL VALUE
HK\$1,110,424.11

TOTAL PNL
HK\$14,834.42 (1.34%)

REALIZED PNL
HK\$221.91 (0.02%)

UNREALIZED PNL
HK\$14,612.51 (1.32%)

IBKR Connected

IBKR

Alpaca

0 BOT POWER

On Running

Strategy Engine Healthy

Open 1076 req / 0 av / 1100.0ms avg

Searching market for opportunities. 3m 5s ago

Active data source: IBKR

HomeTrade DeskPortfolioWatchlistsSim PortfoliosStrategiesToolsForex Desk

Crypto DeskResearchOperationsBot Configuration

No open positions.

IERG4340 Trading Bot

Multi-strategy paper trading dashboard

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Crypto DeskResearchOperationsBot Configuration

Trading LogBot Activity

EXECUTION LEDGER

Trading Log

Spreadsheet view of executed trades, with focus on options and forex activity.

All AssetsAll StatusAll ExecutionSearch symbol / tra

21 recordsSource: COMBINED (LEDGER + BROKER)

CSVExcel

This is all paper mode. Bot Ledger = records written by this bot. Broker Fill Feed = executions reported by IB fill feed.

Time (HKT)	Trade ID	Execution	Type	Symbol	Side	Qty	Entry	Exit	Fees	Net PNL	Strategy
14/04/2025, 23:45:14 HKT	GRS-20250414234374531400	BOT LEDGER	CRYPTO	BTC-USD	BUY	0.003	\$76,064.96	\$76,372.67	\$15.04	+\$30.97	crypto_
14/04/2025, 23:59:00 HKT	GRS-20250414233330778619	BOT LEDGER	CRYPTO	BTC-USD	BUY	0.10	\$74,775.15	\$75,064.96	\$14.20	+\$40.74	crypto_

Quick Demo Flow (90 seconds)

1. Home pulse

Show total PnL, broker state, and active data source in one glance.

2. Trade Desk signals

Open candidates, then switch to Strategy tab to show configurable logic.

3. Risk gate

Move to Risk tab and explain drawdown-aware controls before execution.

4. Traceability

Open Operations log and Bot Activity to prove monitoring and auditability.

What We Improve Next

IN-APP MANUAL ORDER TICKET

Place manual orders inside this app instead of broker software, so broker API sessions can stay connected continuously.

EXECUTION SAFEGUARDS

Add pre-trade checks: max slippage, notional limits, and hard kill-switch with one tap.

REGIME INTELLIGENCE

Expand macro-aware allocation and auto-adjust strategy weights by market context.

Thank You

Live demo and Q&A

IERG4340 Paper Trading Bot

References (IEEE)

- [1] Bank for International Settlements, "OTC foreign exchange turnover in April 2022," Triennial Central Bank Survey, Oct. 27, 2022. [Online]. Available: https://www.bis.org/statistics/rpfx22_fx.htm. [Accessed: Apr. 14, 2026].
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- [3] United Nations, "Population," Global Issues, 2026. [Online]. Available: <https://www.un.org/en/global-issues/population>. [Accessed: Apr. 14, 2026].
- [4] Gallup, "What Percentage of Americans Own Stock?," May 5, 2025. [Online]. Available: <https://news.gallup.com/poll/266807/percentage-americans-owns-stock.aspx>. [Accessed: Apr. 14, 2026].